

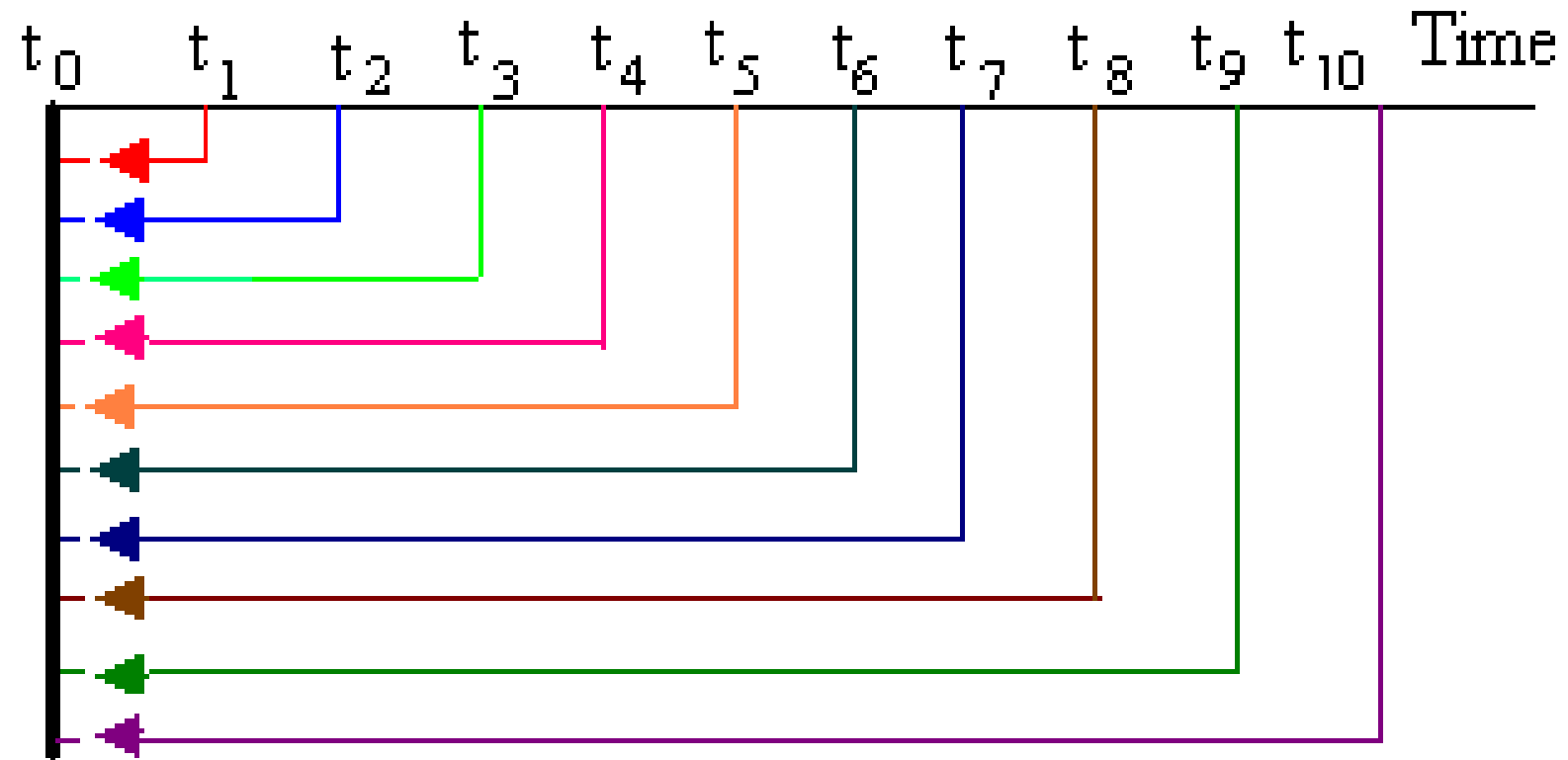
Corporate Finance

Time Value

- Concept of time value is applicable to all aspects of financial management decisions
- An important principle in financial management is that the value of money depends on when the cash flow occurs – INR 100 now is worth more than INR 100 at some future time.
- In organizations, flow of money occurs at various points of time. In order to evaluate the worth of money the financial managers need to look at it from a common platform, namely one time duration.

Comparison over Time

Common Platform for Equating Monetary Flows



Monetary Value

- Money like any other desirable commodity has a price. If you own money, you can, 'rent' it to someone else, say a banker, who can use it to earn income. This 'rent' is usually in the form of interest. The investor's return, which reflects the time-value of money, therefore indicates that there are investment opportunities available in the market.
- Risk-free rate of return rewarding investors for forgoing immediate consumption
- Compensation for risk and loss of purchasing power.

Economic Rationale

- **Risk:** An amount of INR 100 now is certain, whereas INR 100 receivable next year is less certain. This 'uncertainty' principle affects many aspects of financial management and is termed as risk value of money.
- **Inflation:** Under inflationary conditions, the value of money, expressed in terms of its purchasing power over goods and services, declines. Hence INR 100 possessed now is not equivalent to INR 100 to be received in the future.
- **Personal consumption preference:** Most of us have a strong preference for immediate rather than delayed consumption. As a result we tend to value the INR 100 to be received now more than INR 100 to be received latter.

Corporate Decisions

- Consumption decisions: How much of available resources should be spend on immediate consumption?
- Investment decisions: How much of the resources available should be forgone now in the expectation of increased resources at some time in the future? How should such decisions be made?
- Financing decisions: How much cash should be borrowed or lent to enable the above investment and consumption decisions?

Note: Clearly, these decisions are interrelated and should not, therefore, be viewed in isolation.

Goals of Financial Management

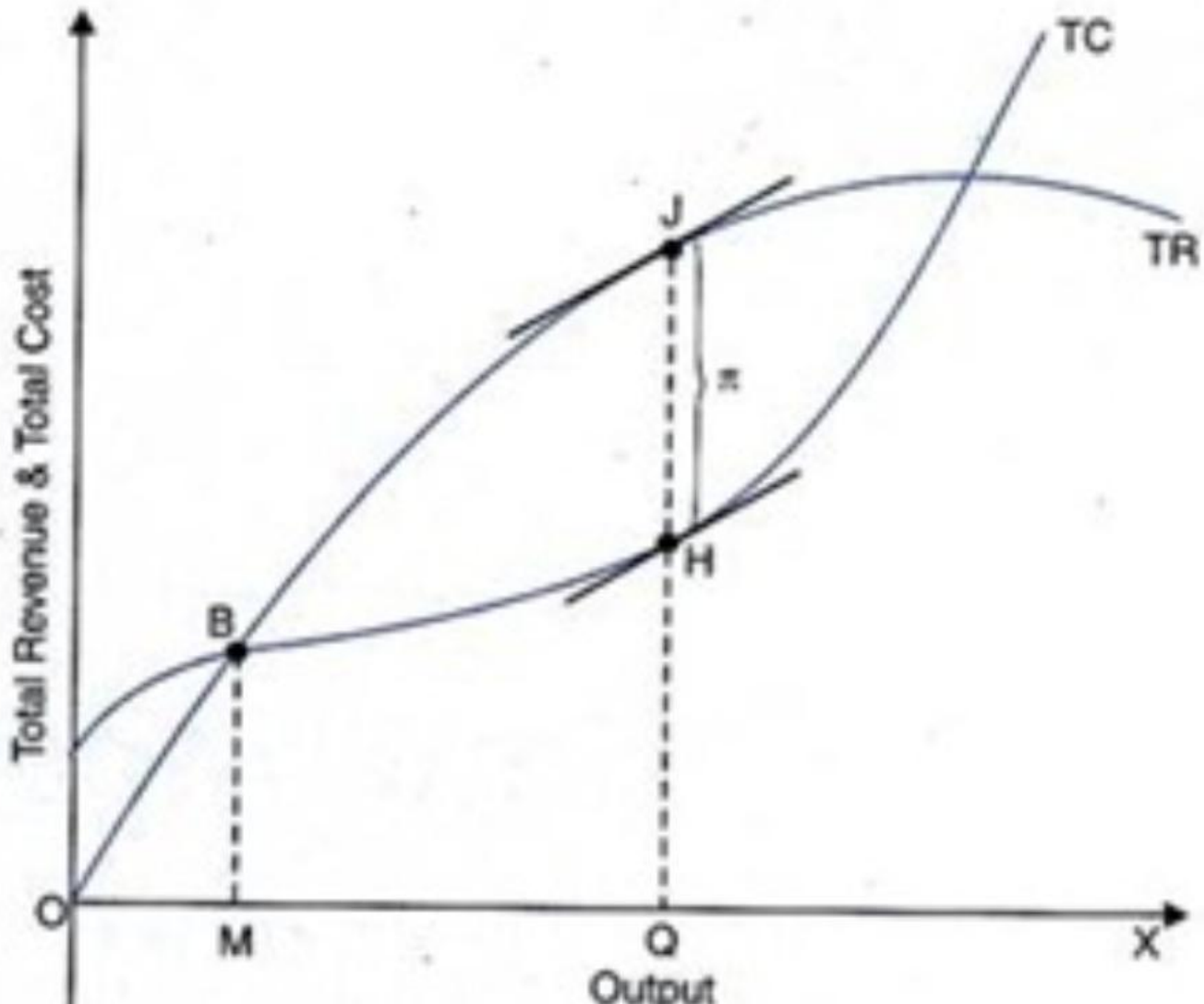
- Profit Maximization
 - Economic profits is the difference between total revenue and economic costs.
- Wealth Maximization
 - Managerial investment decisions vs. cash distribution decisions.
- Value Maximization
 - Value of the firm is the expected present value of future net cash flows discounted to the present.
 - Present value of future profits.

A Comparison of Finance Goals

- When time is short with little uncertainty, profit maximization and wealth maximization are similar.
- Share price represents owner's wealth in a firm, hence addressing share price optimization is similar to wealth maximization objective.
- Firms are expected to forego their short term profits (short term profit maximization objective) to increase long term profits. Value maximization hence addresses the wealth maximization objective.

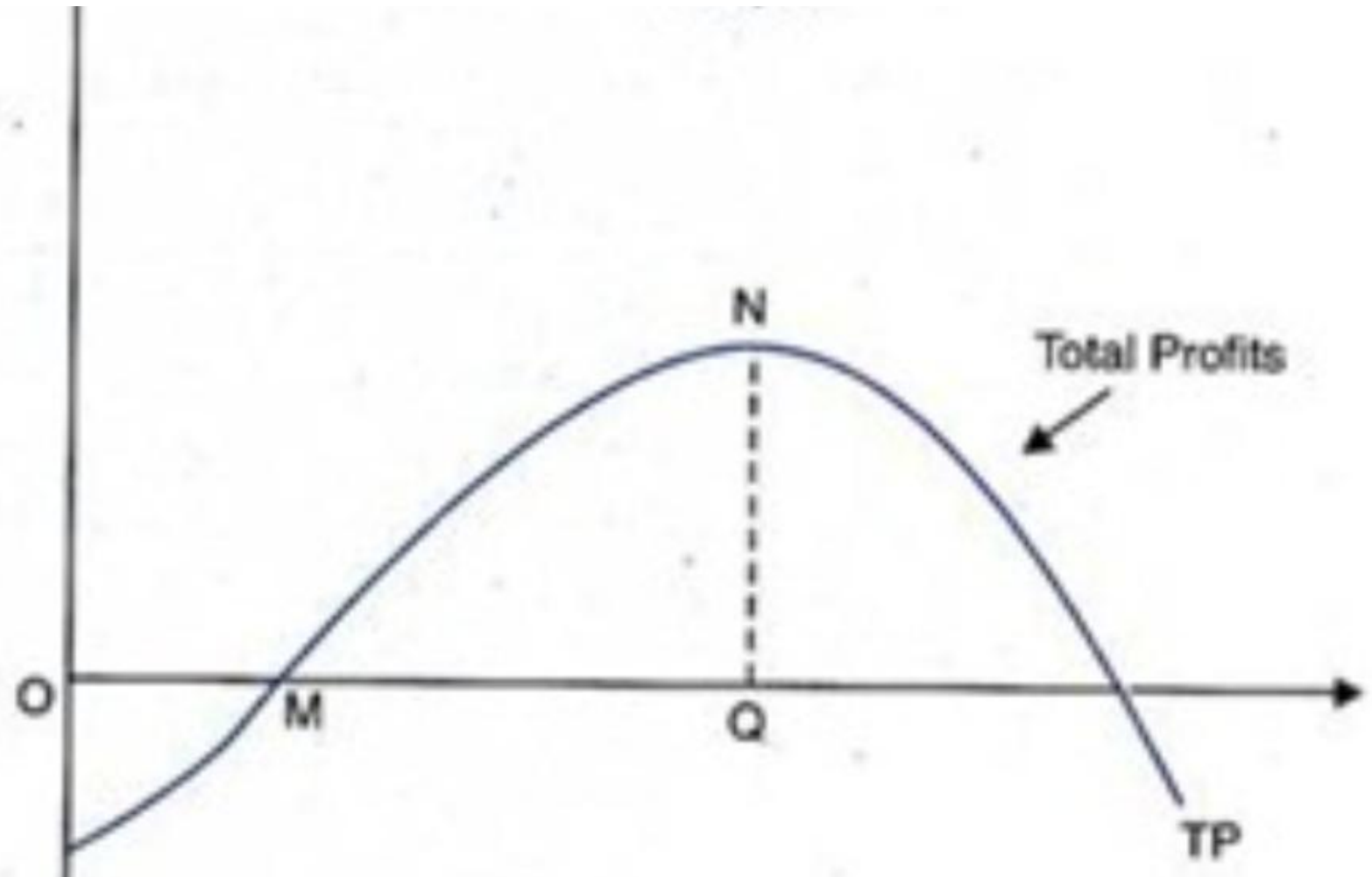
Profit Maximization

- Profits = $TR - TC$
- At OM level of output, total revenue equals total economic costs (breaking even).
- Profits increase till output level OQ is reached.
- At output OQ, difference between total revenue and total cost is maximum. JH is the largest distance between the TR and TC curves.



Total Profit Maximization

- Total profit curve TP which first rises and then beyond point N (corresponding to output level OQ) it starts falling indicating that profits are maximum at output level OQ.
- Price or average revenue equals total revenue divided by a level of output, price charged by the firm at output level OQ is TR/OQ or QJ/OQ



Wealth Maximization

- Choice of how much of wealth should be consumed immediately, and how much should be invested for consumption at a later date.
- Cash generated from business operations can be utilized in two ways:
 - Distributed to the shareholders in the form of a dividend, or reinvested within the business.
- Directors decide how much of the shareholders' wealth to distribute in the form of dividends and how much to withhold for investment purposes, such as building up stock levels or purchasing new equipment.
- Shareholders will only be willing to forgo a higher present level of consumption (in the form of dividends) if they expect an even greater future level of consumption. It is this willingness to give up consumption now with the aim of increasing future consumption which characterizes investment financing decisions.

Investment Consumption Decision Model

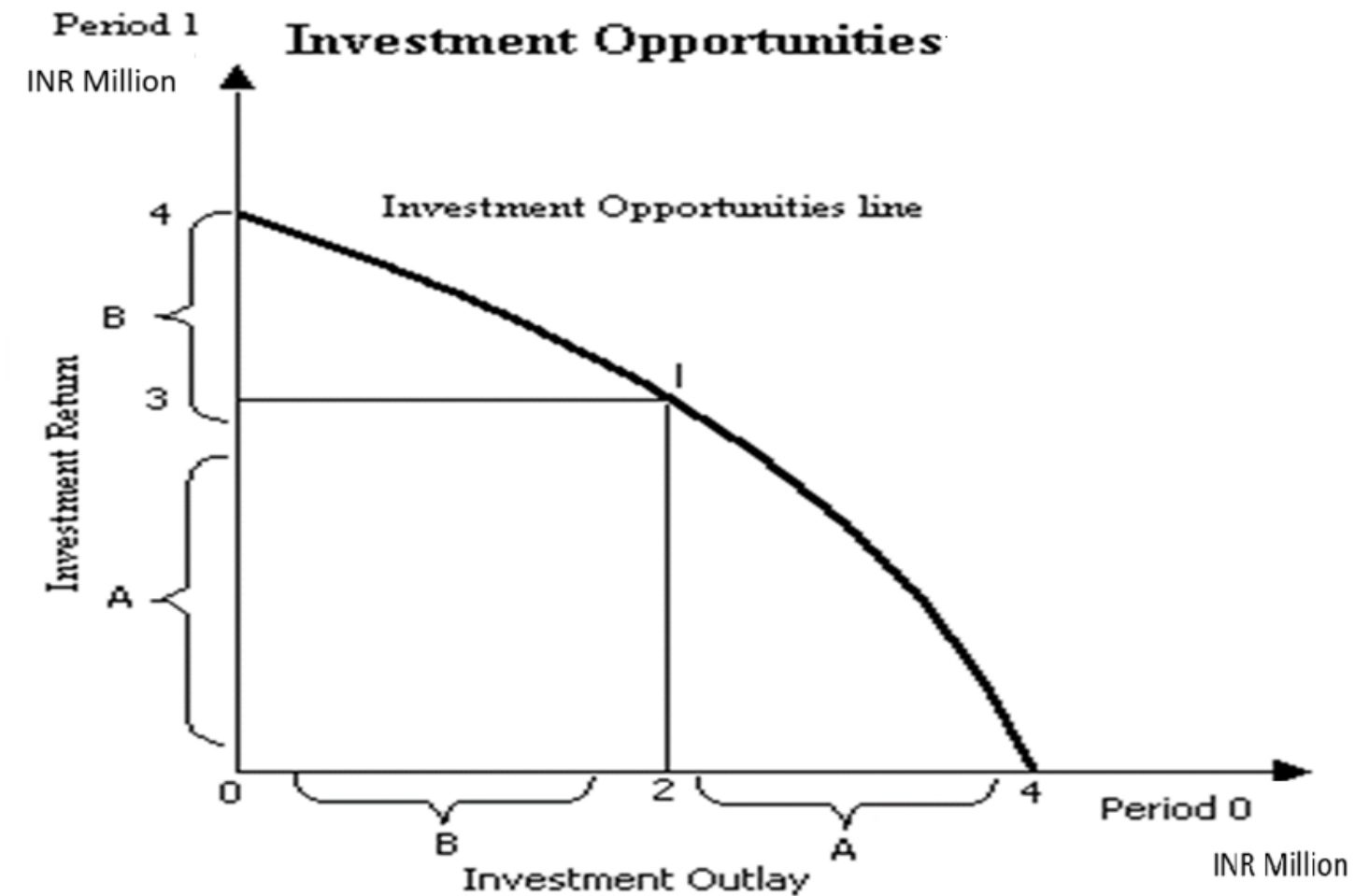
- A new business has just one asset worth INR 4 million.
- **Investors are wealth-maximizers.**
- **Only two periods are considered:** Present period (t_0) and the next period (t_1). Two-period model implies that investments involve an immediate cash outlay in order to receive a return in terms of cash benefit in the following period, t_1 .
- Assume that for investment there are only two possible projects, each costing INR 2 million and having a one-year life.

Investment Opportunity Line

- Investment opportunities line is concave to the origin rather than a straight line. This shape indicates the decreasing returns to scale of each subsequent investment opportunity.
- As a wealth-maximizer, business will first select those investment projects offering the greatest return and work down towards those offering the least return.
- In this investment opportunity line, the business will stop investment.
- Marginal return from the next INR 1 in investment would not be sufficient to compensate for the sacrifice involved in giving up a further INR 1 in dividends.
- Investment decision point is where the marginal return on investment equals marginal rate of time preference.

Investment Opportunity Curve

- Invest INR 4 million, in both projects, for INR 4 million pay-off next year.
- Preference to invest only in project A, costing INR 2 million, that is giving a pay-off of INR 3 million.
- Project B is unprofitable, offering only INR 1 million from INR 2 million investment.
- Point 'I' represents the marginal project beyond which it ceases to be worthwhile to invest.



Return Expectations

- If there are no opportunities to invest surplus cash externally, (one-year deposit or investing in short-term securities), business would have to pay a dividend to shareholders of the INR 2 million unused cash.
- Decision between cash today and cash receivable next year. In effect, business requires a rate of exchange for the transfer of wealth across time.
- Suppose business requires a minimum of INR 115 receivable next year to induce to give up INR 100 now, the rate of exchange would be INR 115(t0): INR 100(t1) or INR 1.15: INR 1. This represents a premium for delayed consumption of one year
 $(\text{INR } 115 / \text{INR } 100) - 1 = 0.15$ or 15%

Expected Rate

- The equivalent rate between today's money and tomorrow's money varies with the level of present consumption sacrificed.
- Business may be willing to forgo the first INR 100 of potential dividend in return for an additional 15 per cent next year, but to persuade to delay the consumption of a further INR 100 will probably require something in excess of 15 per cent. This variable rate for the transfer of wealth across time at various levels of investment is termed the marginal rate of time preference, and will differ for each business.

Managerial Insights

- Managers are assumed to act in the best interest of the owners or shareholders. This they can do by seeking to increase shareholders' wealth in the form of maximizing cash flows through time. There is a market rate of exchange between current and future wealth which is reflected in the current rate of interest.
- Managers should undertake all projects up to the point at which the marginal return on the investment is equal to the rate of interest on equivalent financial investments in the capital market. The result is an increase in the market value of the firm and thus in the market value of the shareholders' stake in the firm.
- Management need not concern itself with shareholders' particular time patterns of consumption or risk preferences. In well-functioning capital markets, shareholders can borrow or lend funds to achieve their personal consumption requirements. Furthermore, by carefully combining risky and safe investments they can achieve the desired risk characteristics for those consumption requirements.